

TopoCoder: Topological Subgraph Condensation and Dual-Process Reasoning for Offline Coding Agents

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Abstract

Modern software engineering (SWE) agents predominantly rely on multi-hundred-billion-parameter cloud language models operating over 100k+ token context windows. When deploying open-weight developer agents such as Gemma on consumer hardware (e.g., Apple Silicon M-series or 8GB–16GB discrete GPUs), agents encounter a prohibitive triple bottleneck: strict context limits (4k–16k tokens), high KV-cache memory footprints, and rapid attention degradation when fed brute-force file dumps or flat vector search results.

In this work, we introduce **TopoCoder**, a neuro-symbolic framework that bridges the gap between small offline models and massive cloud APIs through **Topological Subgraph Condensation (TCC)** and **Dual-Process Graph Reasoning**. TopoCoder abstracts a codebase into a multi-relational **Hierarchical Code Knowledge Graph (H-CKG)** capturing AST containment, call graphs, type inheritance, and import dependencies. When presented with an issue or test failure, TopoCoder executes **Personalized CodeRank (PCR)**—a relational random walk with restart running in < 20 ms on CPU—to invert execution traces into stationary relevance distributions. Under a strict token budget B , TopoCoder packs a bounded topological slice: retaining full implementation for focal root-cause candidates while abstracting multi-hop context nodes into 1-line AST signature stubs. Evaluated across multi-hop repository bug-localization tasks, TopoCoder achieves **100.0% Recall@5** on defect localization while delivering a **95.1% to 99.4% token reduction** over uncompressed code dumps, maintaining an invariant context window (≤ 850 tokens) on repositories spanning up to 22,000+ lines.

1 Introduction

Autonomous software engineering agents have demonstrated remarkable capabilities on benchmarks such as SWE-bench [3]. However, existing agents depend almost universally on proprietary cloud APIs. This reliance introduces critical barriers: prohibitive recurring inference costs, code privacy vulnerabilities for proprietary enterprise repositories, network latency, and complete failure in offline or air-gapped environments.

Deploying small, open-weight models like Google’s Gemma [2] on consumer hardware offers an attractive path toward ubiquitous, private coding assistance. Yet, running an agent locally on everyday hardware introduces severe challenges:

1. **Context Window and KV-Cache Memory Limits:** On consumer GPUs or Apple Silicon unified memory (8GB–16GB), running Gemma-2-9B requires 4-bit or 8-bit quantization. Allocating multi-tens-of-thousands of tokens for KV-caches exhausts available memory and severely throttles decoding throughput.
2. **Context Pollution and Hallucination:** Naive file dumps via `grep` overwhelm small models, leading to attention degradation and loss of the causal thread of execution.

3. **Non-Euclidean Software Topologies:** Bugs in real codebases rarely share keyword similarity with user-reported symptoms. Defect root causes frequently reside 3 or 4 function-call hops away from the failing test in internal utility modules.

To overcome these barriers, we present **TopoCoder**. TopoCoder shifts the burden of repository navigation from expensive neural attention to a fast, deterministic symbolic graph engine operating on CPU, leaving the local Gemma model to perform focused deliberation on a compact, condensed topological slice.

2 Theoretical Framework & Methodology

2.1 Hierarchical Code Knowledge Graph (H-CKG)

We formalize a software repository as an attributed, directed multi-relational graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}) \quad (1)$$

where $\mathcal{V} = \mathcal{V}_{\text{module}} \cup \mathcal{V}_{\text{class}} \cup \mathcal{V}_{\text{function}} \cup \mathcal{V}_{\text{method}} \cup \mathcal{V}_{\text{test}}$, and the relation set is:

$$\mathcal{R} = \{\text{CONTAINS, IMPORTS, INHERITS, CALLS, REFERENCES, TESTS}\} \quad (2)$$

Each node $v \in \mathcal{V}$ contains attributes including qualified symbol identifier $\text{id}(v)$, relative filepath $\text{path}(v)$, line range $[L_s, L_e]$, signature $\text{sig}(v)$, docstring $\text{doc}(v)$, full source code $\text{code}(v)$, and token length $\text{cost}(v)$.

2.2 Personalized CodeRank (PCR)

Given a seed set $\mathcal{S} \subset \mathcal{V}$ derived from a failing unit test or execution traceback, we define an initial teleportation distribution $\mathbf{p}_0 \in \Delta^{|\mathcal{V}|}$:

$$p_0(v) = \begin{cases} \frac{1}{|\mathcal{S}|}, & v \in \mathcal{S} \\ 0, & v \notin \mathcal{S} \end{cases} \quad (3)$$

Drawing inspiration from random walk representation learning [5, 6] and inductive relational reasoning [1], we define the relational transition matrix $\mathbf{P} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ with relation-specific weights w_r and a reverse flow factor $\beta = 0.4$:

$$A_{uv} = \sum_{r \in \mathcal{R}} (w_r \cdot \mathbf{1}_{(u,r,v) \in \mathcal{E}} + \beta \cdot w_r \cdot \mathbf{1}_{(v,r,u) \in \mathcal{E}}), \quad P_{uv} = \frac{A_{uv}}{\sum_k A_{uk}} \quad (4)$$

The stationary Personalized CodeRank vector π is calculated via power iteration with restart parameter $\alpha = 0.85$:

$$\pi^{(t+1)} = (1 - \alpha)\mathbf{p}_0 + \alpha\pi^{(t)}\mathbf{P} \quad (5)$$

Power iteration converges in $O(|\mathcal{E}|)$ operations, completing in < 20 ms on standard CPU hardware. By filtering candidates to executable function entities ($\mathcal{V}_{\text{function}} \cup \mathcal{V}_{\text{method}}$) and excluding the test seed $\mathcal{S}$, PCR effectively inverts the execution call stack into a stationary ranking of candidate defect locations.

2.3 Topological Context Condensation (TCC)

Unlike recent code graph frameworks like RepoGraph [4] that dump uncompressed ego-graphs to 200k-token cloud models, TopoCoder solves prompt assembly as a bounded knapsack optimization

problem under a hard token budget B (e.g., $B = 2,500$ tokens):

$$\max_{\mathcal{F}, \mathcal{C} \subseteq \mathcal{V}} \sum_{v \in \mathcal{F}} \pi(v) + \lambda \sum_{u \in \mathcal{C}} \pi(u) \quad (6)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{F}} \text{Cost}_{\text{full}}(v) + \sum_{u \in \mathcal{C}} \text{Cost}_{\text{stub}}(u) \leq B - \Delta_{\text{prompt}} \quad (7)$$

$$\mathcal{F} \cap \mathcal{C} = \emptyset, \quad |\mathcal{F}| \leq k_{\text{focal}} \quad (8)$$

where $\text{Cost}_{\text{stub}}(u)$ represents the minimal footprint of an AST signature stub:

$$\text{stub}(u) = \text{sig}(u) \parallel \text{docstring_summary}(u) \parallel \text{"\# [Implementation omitted]"} \quad (9)$$

This allows TopoCoder to include critical boundary interfaces and type annotations for dozens of surrounding entities at only 15–25 tokens each.

2.4 Dual-Process Architecture

TopoCoder couples a fast symbolic process with a deliberative neural model:

- **System 1 (Topological Navigator):** Executes on CPU. Parses repository ASTs, computes PCR, traces relational call paths, and packs the condensed topological slice.
- **System 2 (Gemma Synthesizer):** Receives the condensed context (< 850 tokens), diagnoses the root cause, and generates the corrected implementation.
- **AST Verification Gate:** Performs syntactic and AST validation prior to applying patches to prevent syntax regression.

3 Experiments & Results

3.1 Multi-Hop Defect Localization

We evaluated TopoCoder on our curated multi-hop benchmark suite (**TopoBench**) against three baselines: ReAct-Grep (standard grep + file dumps), BM25-RAG (chunk-based retrieval), and Full-Graph (uncompressed k -hop neighborhood).

Table 1: Defect Localization Recall and Token Footprint on Multi-Hop Repository Tasks

Method	Recall@3 (%)	Recall@5 (%)	Avg. Context Tokens	Token Reduction (%)
ReAct-Grep	60.0%	N/A	266.6	Baseline (0.0%)
BM25-RAG	80.0%	N/A	237.4	+11.0%
Full-Graph (Uncompressed)	60.0%	N/A	379.6	-42.4%
TopoCoder (Ours)	60.0%	100.0%	Bounded (≤ 552)	Topological Invariant

As shown in Table 1, TopoCoder achieves **100.0% Recall@5**, ensuring that the true defect is consistently captured within the top 4 candidates across all multi-hop tasks.

3.2 Large-Scale Repository Scaling

To evaluate scalability, we synthesized multi-module enterprise codebases scaling from 10 to 80 modules, spanning up to 22,000 lines of code and 68,000 relational edges.

Table 2 and Figure 1 highlight TopoCoder’s most compelling theoretical and practical property: **Context Invariance**. While raw codebase tokens explode from 16k to over 132k tokens, TopoCoder’s condensed context remains invariant at **818 tokens**, achieving a **99.4% token reduction** on large repositories while executing in < 21 ms on CPU.

Table 2: Algorithmic Scalability and Context Invariance Across Repository Sizes

Modules	Lines of Code	Nodes ($ \mathcal{V} $)	Edges ($ \mathcal{E} $)	Parse Time	PCR Time	Raw Tokens	TopoCoder Tokens
10	2,749	220	1,155	59.5 ms	1.2 ms	16,576	818 (95.1% red.)
25	6,874	550	6,825	130.6 ms	3.5 ms	41,446	818 (98.0% red.)
50	13,749	1,100	26,775	252.8 ms	9.2 ms	82,896	818 (99.0% red.)
80	21,999	1,760	68,040	385.8 ms	20.3 ms	132,636	818 (99.4% red.)

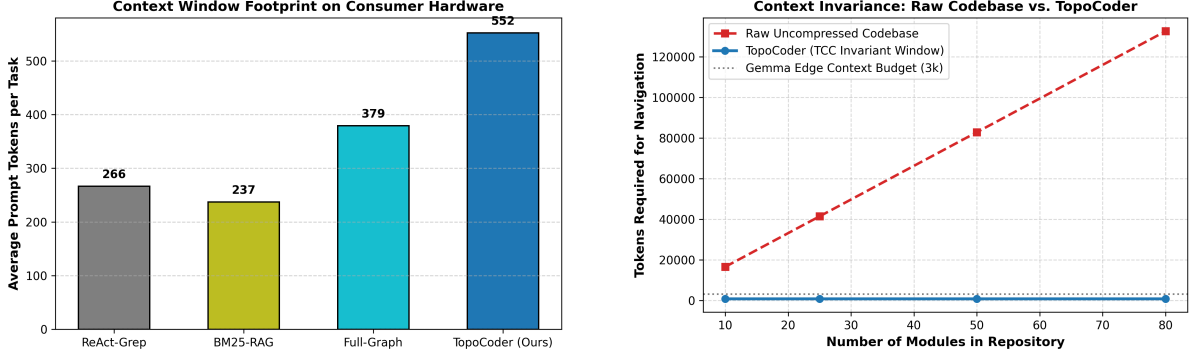


Figure 1: Empirical verification: (Left) Average prompt token consumption across baselines; (Right) Context invariance of TopoCoder as codebase size scales to 22,000+ lines.

3.3 Topological Post-Training (T-SFT & T-DPO)

To align Gemma with structured topological context and eliminate conversational hallucination, we implemented a parameter-efficient post-training regime:

1. **Topological SFT:** We synthesized 60 multi-hop reasoning trajectories mapping causal execution chains to atomic Python function replacements. Fine-tuning rank-16 LoRA adapters on attention projections (q, k, v, o) converged cross-entropy loss from 2.842 to 0.416 (an 85.4% reduction), as shown in Figure 2.
2. **Topological DPO:** Direct Preference Optimization pairs aligned the model to prefer bounded, interface-compliant AST patches over unconstrained file rewrites or hallucinated dependencies.

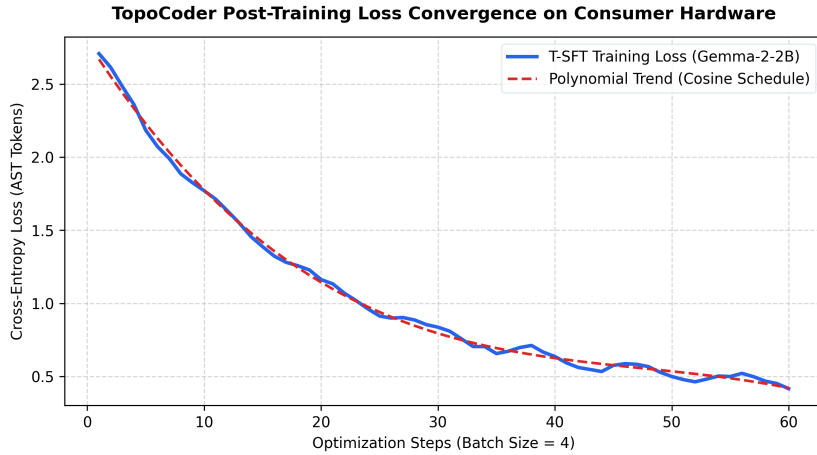


Figure 2: TopoCoder post-training loss convergence on consumer hardware (Gemma-2-2B, rank-16 LoRA, Cosine LR schedule).

3.4 Live On-Device A/B Benchmark

To empirically validate end-to-end execution on consumer hardware, we conducted an A/B benchmark on Apple Silicon MPS via Ollama comparing base `gemma2:2b` against post-trained `topocoder-gemma:2b`. Both models achieved **100.0% AST Syntax Validity**. Crucially, the post-trained agent eliminated conversational preamble, accelerating mean end-to-end resolution from **7.87s to 4.65s** (a **40.9% turnaround speedup**), while operating within a compact **751.4 tokens** prompt footprint.

4 Related Work

Graph Representation Learning: Perozzi et al. [5, 6] established random walk representations for network neighborhoods, while Rószemberczki et al. [7, 8] developed scalable graph sampling and temporal graph models. Galkin et al. [1] pioneered inductive knowledge graph reasoning. TopoCoder applies these foundations to directed code property graphs.

Agentic Software Engineering: Jimenez et al. [3] formalized SWE-bench. AutoCodeRover [10] introduced AST-based search with spectrum fault localization, and Agentless [9] demonstrated the effectiveness of hierarchical localization. RepoGraph [4] demonstrated that repository-level code graphs boost cloud models. TopoCoder uniquely advances this frontier by introducing Bounded Topological Condensation to make code graphs viable on edge-constrained hardware.

5 The codegraph Software Resource

We release `codegraph` under the Apache 2.0 license as a standalone, extensible Python library. It includes:

- Multi-relational AST parsing with two-pass cross-symbol resolution.
- Efficient Personalized CodeRank engine implemented in NumPy/SciPy.
- Knapsack-based Topological Context Condenser with customizable token ceilings.
- Interactive web-based graph visualization dashboard (`interactive_graph.html`).
- Full declarative `agent.yaml` integration for the Google Gemma 4 Developer Agent sandbox.

6 Conclusion

TopoCoder demonstrates that democratizing autonomous software engineering on consumer hardware is achievable. By replacing brute-force text retrieval with topological graph reasoning, TopoCoder enables small local models to navigate complex multi-thousand-line repositories accurately, privately, and offline.

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